

END SEMESTER ASSESSMENT (ESA) - JULY - 2023**UE20CS351 - Cloud Computing****Total Marks : 100.0**

1.a. Explain the following terms with respect to Cloud Computing

1. On-demand service
2. Rapid provisioning
3. Measured Service
4. Resource Pooling
5. Availability
6. Broad Network Access

(6.0 Marks)

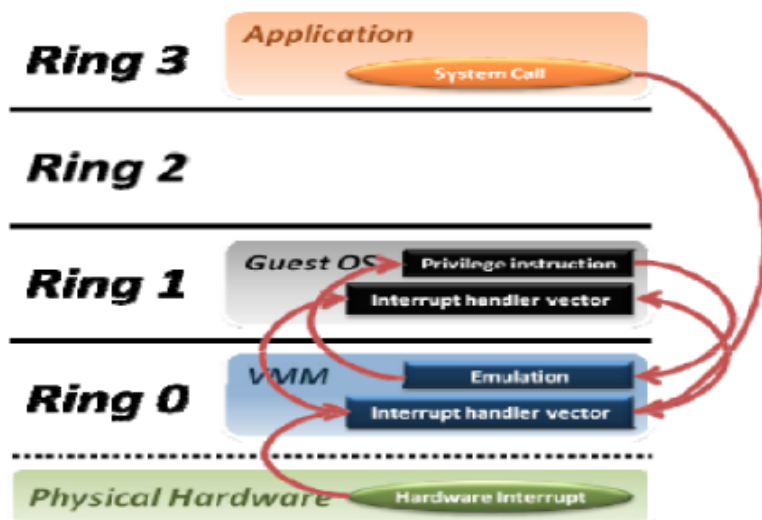
1.b. What is elasticity? How is it different from scalability? Give an example each to clearly distinguish these concepts, **(4.0 Marks)**

1.c. What are private, public, and hybrid clouds? Explain each of them. Discuss the relative advantages and disadvantages. **(10.0 Marks)**

2.a. Distinguish between bare metal and hosted hypervisors. Give an example of each. (4.0 Marks)

2.b. What is DevOps? What are the benefits derived from DevOps? How is DevOps different from traditional software development and Operations processes? (4.0 Marks)

2.c. Explain what are rings. Explain the instruction executions depicted in the following image. (8.0 Marks)



2.d. List at least four similarities and differences between Containers and VMs.
(4.0 Marks)

3.a. Explain what Gluster and Lustre are. Explain the Gluster architecture. How is it different from Lustre?
(5.0 Marks)

3.b. What is a consistency model? Explain Strict, Sequential, Causal and PRAM consistency.
(9.0 Marks)

3.c. State CAP theorem. Discuss 3 desirable properties of the CAP theorem and some of its practical implications.
(6.0 Marks)

4.a. What is the purpose of Leader Election in Distributed computing? Explain briefly Bully Algorithm and Modified Ring election. (10.0 Marks)

4.b. Explain how Zookeeper works. Explain the key benefits and the common services offered by Zookeeper. (10.0 Marks)

5.a. Explain the terms mentioned below from a cloud security perspective

1. Cloud Time Service
2. Identity Management
3. Access Management
4. Break Glass Procedures
5. Key Management

(10.0 Marks)

5.b. What is multi-tenancy? Mention its benefits in Cloud Computing. You are asked to design a multitenant database for two universities – HighTechUniv and GlobalUniv to store information about students. HighTechUniv wants to store USN, student names and email ids while GlobalUniv wants to store USN, student names, and grades. Design a multitenant database using the preallocated column method for the same. (6.0 Marks)

5.c. Explain the following security design patterns

a. Defense in Depth

b. Honeypots

(4.0 Marks)